Ceiling Diffuser With Double Deflection CD-DD



Description
Aluminum rectangular diffuser designed for paneled ceiling mounting.

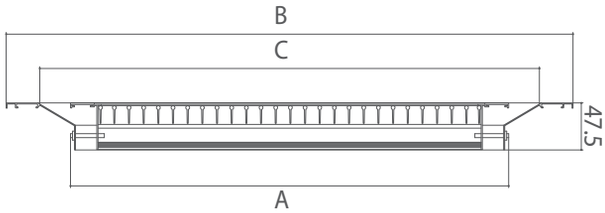
Application
Suitable for supply and exhaust air systems.

Characteristics
The diffuser is provided with a central core that consists of a **DD** -grille (double deflection) with two rows of individually adjustable blades.
The adjustable baldes allow airflow distribution to desired areas.
Removable central cone for easy installation and cleaning.
The air flow can be adjusted by mounting an air control damper **OBD** on the diffuser inlet connection.

Accessories
Air control damper with opposable blades **OBD**
Insulated/Non-insulated adapter **ADP IZ/N**

Materials and finishing
Extruded aluminum material, electrostatic field painted white RAL 9016
Other RAL colors available on request
Natural anodised aluminum

Technical design



Assembly Methods
Paneled ceiling 600x600mm (standard)
Mounting holes on the diffuser inlet connection (optional)

Selection table CD-DD

Air flow m³/h	LXH	150x150	225x225	300x300	375x375	455x455	480x480	525x525	600x600
	Ak [m²]	0.014	0.028	0.048	0.075	0.110	0.123	0.147	0.192
100	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]	2.0 3.0 20.0 3.0							
160	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]	3.2 3.7 23.0 7.0	1.6 2.4 20.0 2.0						
200	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]	4.0 5.0 25.0 12.0	2.0 3.5 20.0 3.0						
300	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]	6.0 7.5 30.0 30.0	3.0 5.3 25.0 7.0	1.7 2.5 20.0 3.0					
400	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]		4.0 7.0 30.0 12.0	2.3 4.0 23.0 5.0	1.5 2.0 15.0 2.0				
500	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]		4.9 9.0 30.0 20.0	2.9 5.0 28.0 7.0	1.9 1.3 19.0 3.0				
600	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]		5.9 9.0 35.0 33.0	3.4 7.0 33.0 9.0	2.2 4.0 25.0 4.0	1.5 3.0 17.0 2.0			
700	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]		6.9 12.0 40.0 40.0	4.0 8.0 35.0 15.0	2.6 7.5 30.0 5.0	1.8 5.0 21.0 3.0			
800	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]		7.9 18.0 45.0 50.0	4.6 11.0 40.0 20.0	3.0 8.0 31.0 7.0	2.0 6.0 23.0 4.0	1.8 3.0 20.0 3.0	1.5 2.0 17.0 2.0	
900	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]			5.2 12.0 43.0 25.0	3.3 9.0 34.0 15.0	2.3 7.0 26.0 5.0	2.0 4.0 23.0 3.0	1.7 2.5 20.0 2.0	
1000	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]			5.7 15.0 46.0 32.0	3.7 12.0 36.0 18.0	2.5 8.0 29.0 6.0	2.3 7.0 25.0 4.0	1.9 4.0 23.0 3.0	1.4 2.0 18.0 2.0
1200	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]			6.9 20.0 50.0 41.0	4.4 15.0 41.0 20.0	3.0 11.0 33.0 8.0	2.7 9.0 30.0 6.0	2.3 7.0 27.0 4.0	1.7 3.0 22.0 3.0
1600	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]				5.9 17.0 48.0 29.0	4.0 13.0 40.0 14.0	3.6 12.0 37.0 10.0	3.0 10.0 33.0 8.0	2.3 7.0 29.0 5.0
2000	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]				7.4 20.0 53.0 45.0	5.0 17.0 45.0 23.0	4.5 15.0 42.0 18.0	3.8 11.0 40.0 12.0	2.9 8.5 34.0 7.0
3000	Veff [m/s] X (m) NR [dB(A)] ΔPt [pa]					7.6 20.0 55.0 50.0	6.8 18.0 53.0 40.0	5.7 15.0 50.0 25.0	4.3 12.0 44.0 16.0

Annotation

* Ak [m²] - Free surface of the diffuser
* Veff[m/s] - Air velocity in diffuser
* X [m] - Air jet length
* ΔPt [pa] - Pressure loss

* NR [dB(A)] - Noise level without the room attenuation

Standard dimensions

CD - DD			
neck/external size	**A**	**B**	**C**
150 / 290	150 x 150	290 x 290	230 x 230
150 / 595	150 x 150	595 x 595	235 x 235
225 / 365	225 x 225	365 x 365	305 x 305
225 / 595	225 x 225	595 x 595	315 x 315
300 / 440	300 x 300	440 x 440	380 x 380
300 / 595	300 x 300	595 x 595	385 x 385
375 / 515	375 x 375	515 x 515	455 x 455
375 / 595	375 x 375	595 x 595	465 x 465
455 / 595	455 x 455	595 x 595	535 x 535
480 / 620	480 x 480	620 x 620	560 x 560
525 / 665	525 x 525	665 x 665	605 x 605
600 / 740	600 x 600	740 x 740	680 x 680

Order codes

Type + **OBD** + **ADP IZ/N** / **Dimensions** / **L X H** / **O**

- **Type**: **CD-DD**
- **OBD**: Air control damper
- **ADP IZ/N**: Insulated/Non-insulated adapter
- **Dimensions**: **neck/external size**
  150 / 290
  150 / 595
  225 / 365
  225 / 595
  300 / 440
  375 / 515
  375 / 595
  455 / 595
  480 / 620
  525 / 665
  600 / 740
- **L X H**: Other dimensions on request
- **O**: **0** Painted white RAL 9016
  R Other RAL paints
  Natural anodised